

Football Match Manager - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager API provides RESTful endpoints for managing football matches, including authentication, match retrieval, and user follow functionality. The API utilizes JWT tokens for authentication and implements rate limiting for both auth and other endpoints. The API is versioned, with breaking changes incrementing the version number.

1.2 Maturity Assessment

The API specification is mostly complete, but there are some gaps identified, including the lack of sections for goals and objectives, functional requirements, non-functional requirements, scope and out-of-scope items, and open questions or uncertainties. The 'NEEDS_REFINEMENT' status reflects the need to address these gaps to ensure the API meets its intended purpose and is properly documented.

1.3 Technical Stack

- Python
- Flask
- JWT

1.4 Architectural Constraints

- Rate limiting for auth endpoints: 5 requests per minute per IP
- Rate limiting for other endpoints: 60 requests per minute per user
- Validation for query parameters (e.g., leagueId, dateFrom, dateTo, status, search)
- Pagination with default page number 1 and default limit 20
- Error handling with HTTP status codes (200, 201, 400, 401, 403, 404, 429, 500)

1.5 Critical Dependencies

- JWT token for authentication
- User entity for follow functionality
- Match entity for retrieval and filtering
- League entity for filtering
- Team entity for match retrieval
- Rate limiting for auth and other endpoints

2. Architecture Workflows & Visual Diagrams

No diagrams provided.

3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Description	Source Section
MATCH-01	Match entity	Match
TEAM-01	Team entity	Team
LEAGUE-01	League entity	League
USER-01	User entity	User (authenticated user only)
REQ-01	Get list of matches with optional filtering	Matches
REQ-02	Get a specific match by ID	Matches
REQ-03	Follow a match	User Follows
NFR-01	Rate limiting for auth endpoints	Rate Limiting
NFR-02	Rate limiting for other endpoints	Rate Limiting

3.2 Security Rules

- Authentication: JWT token required for all endpoints except auth endpoints
- Authorization: User entity required for follow functionality
- Rate limiting: Implemented for both auth and other endpoints

3.3 Data Models

Match

```
{
  "id": "uuid",
  "homeTeam": {
    "id": "uuid",
    "name": "string",
    "abbreviation": "string",
    "crestUrl": "url"
  },
  "awayTeam": {
    "id": "uuid",
    "name": "string",
    "abbreviation": "string",
    "crestUrl": "url"
  },
  "dateTime": "ISO string",
  "venue": "string",
  "league": {
    "id": "uuid",
    "name": "string",
    "country": "string"
  },
  "status": "upcoming|live|finished|postponed|cancelled",
  "homeScore": "integer (nullable)",
  "awayScore": "integer (nullable)",
  "lastUpdated": "ISO string"
}
```

Team

```
{
  "id": "uuid",
  "name": "string",
  "abbreviation": "string",
  "crestUrl": "url",
  "foundedYear": "integer"
}
```

League

```
{
  "id": "uuid",
  "name": "string",
  "sport": "string",
  "country": "string",
  "logoUrl": "url",
  "currentSeason": "string"
}
```

User (authenticated user only)

```
{
  "id": "uuid",
  "email": "string",
  "displayName": "string",
  "avatarUrl": "url"
}
```

4. Project Governance & Structural Gaps

4.1 Structural Gaps

- Missing section: Goals & Objectives (priority: HIGH)
- Missing section: Functional Requirements (priority: MEDIUM)
- Missing section: Non-Functional Requirements (priority: MEDIUM)
- Missing section: Scope & Out-of-Scope (priority: LOW)
- Missing section: Open Questions & Uncertainties (priority: LOW)

4.2 Remediation & Workflow

- Add a section to describe the goals and objectives of the API
- Add a section to describe the functional requirements of the API
- Add a section to describe the non-functional requirements of the API
- Add a section to describe the scope and out-of-scope items for the API
- Add a section to describe any open questions or uncertainties related to the API

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	API Contract: Football Match Manager	The primary interface for interacting with the Football Match Manager backend, providing endpoints for various operations.
CORS Standard	TECHNICAL_STACK	Error Responses	A set of rules for managing cross-origin resource sharing, ensuring secure data exchange between web pages and servers.
ID	BUSINESS_DOMAIN	MATCH-01	A unique identifier assigned to each entity within the system, used for referencing and tracking purposes.
IP	TECHNICAL_STACK	Rate Limiting	A numerical label assigned to each device connected to a computer network, used for communication and identification purposes.

JSON	TECHNICAL_STACK	Match	A lightweight data interchange format, used for exchanging data between web servers and web applications.
JWT	TECHNICAL_STACK	Authentication	An open standard for securely transmitting information between parties, used for authentication and authorization purposes.
UUID	TECHNICAL_STACK	Match	A 128-bit label used for information in computer systems, providing a unique identifier for each entity.
dateFrom	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the start date for the range of matches to be retrieved.
dateTo	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the end date for the range of matches to be retrieved.
leagueId	BUSINESS_DOMAIN	Matches	A parameter used for filtering matches, specifying the league to which the matches belong.